

MONEY CARD — SUPER ADMIN DEMO GUIDE

Comprehensive End-to-End System Walkthrough, Architecture, Malayalam Scripts & FAQs

SUPER ADMIN SCOPE

Multi-Tenant SaaS | Offline POS Sync | Plan Overrides | Real-Time Analytics

1. Executive Overview & Multi-Tenant Architecture

Money Card is an enterprise-grade multi-tenant smart card ecosystem designed for seamless cashless operations in universities, hospital canteens, corporate parks, and multi-branch food courts. The Super Admin holds root authority across the entire platform.

Malayalam Summary / Malayalam Explanation:

Money Card ennal college-ukal, hospital-ukal, IT park-ukal ennivayile cafeteriakale poornamayi cashless aakkunna smart card system aanu. Super Admin aanu platform-inte owner. Super Admin-u keezhil palavidha cafeteriakale (Organizations) register cheyyanum, avarude collection, sales, active cards, plan renewal requests enniva live aayi monitor cheyyanum saadhikkum.

Role Level	Primary Scope & Responsibilities	Key Controls & Privileges
Super Admin	Global Platform Owner (All Cafeterias)	Platform analytics, Plan catalog, Org approvals, Overrides
Org Admin	Single Cafeteria Owner (Multi-Branch)	Staff assignment, card inventory, menu pricing, local reports
Branch Staff	Cashier / POS Operator	Card tap checkout, card recharge, balance inquiries, refunds

2. Super Admin Modules & Core Business Logic

Module 1: Real-Time Platform Dashboard (/)

Central control room featuring Urgent Action Needed alerts, Quick Actions, Time Window filters, and 4 Synchronized KPI Cards (Cafeterias, Sales, Active Cards, Orders). Choosing a cafeteria in the single dropdown immediately synchronizes all 4 KPI boxes to that exact Organization Admin in real time.

Malayalam Explanation: Super Admin login cheyyumpol thanne urgent plan approvals kanikkum. 4 KPI card-ukalil (Cafeteria, Sales, Active Cards, Orders) motham collection live aayi kanam. Oru cafeteria select cheythal aa cafeteria-yude maathram metrics-ilekku taniye maarum.

Key Capabilities: View Cafeterias collapsible directory, Quick action routing, Scoped drill-down.

Module 2: Cafeterias / Organization Management (/organizations)

Onboard new cafeterias, allocate unique organization IDs, assign initial subscription plans, adjust card/staff quota limits, and set status to Active or Suspended based on billing and policy compliance.

Malayalam Explanation: Puthiya cafeteria-kale add cheyyuka, avarude admin email, allocated plan, card limits enniva set cheyyuka, athupole non-payment vannal suspend cheyyuka ennivakkayi upayogikkunnu.

Key Capabilities: Add Organization, Manage Quotas, Set Active/Inactive status.

Module 3: Plans & Subscriptions Management (/subscriptions & /plans)

Streamlined SaaS plan catalog (Create New Plan), pricing intervals (Monthly/Yearly), branch/staff/card quota limits, and one-click review/approval of plan renewal and upgrade requests submitted by Org Admins.

Malayalam Explanation: Create New Plan vazhi subscription tiers undakkuka, quota limits nischayikkuka. Org Admin-mar ayakkunna plan renewal/upgrade requests review cheythu approve cheyyuka.

Key Capabilities: Create New Plan, Review & Approve requests, Status filter, Custom Overrides.

Module 4: Platform Analytics & Client-Side PDF Engine (/analytics)

Financial aggregation of gross purchases, customer recharges, refunds, and net revenue. Includes branch performance matrices and peak demand hour curves. Offers Dual-Mode PDF export: [View PDF] in-browser blob preview & [Download PDF] native vector rendering.

Malayalam Explanation: Motham revenue, recharges, customer refunds, net revenue enniva live chart-ukalil kanam. [View PDF] vazhi browser-il thanne report preview cheyyam, [Download PDF] vazhi direct print-ready file download aakum.

Key Capabilities: Revenue metrics, Peak hour diagnostics, Zero-server-load jsPDF exports.

Module 5: Mobile POS & Scanned Card Hub (Flutter POS Native App)

Cashier POS with single-tap card scan: Priority #1 Recharge Card and #2 Add Products food ordering (row-wise menu). Supports instant order cancellation with card refund, top-up voiding, and role-based analytics.

Malayalam Explanation: Mobile POS-il card scan cheythu udan Recharge (#1), Add Products (#2) cheyyam. Food order cancel cheythu card balance refund nalkam, top-up void cheyyam.

Key Capabilities: Recharge #1, Add Products #2, Order Cancellation & Auto-Refund, Top-up Voiding.

3. Step-by-Step Live Demo Presentation Script

Step 1: Super Admin Login

- English: Log in with super admin credentials. The system recognizes root privileges and directs straight to the platform dashboard.
- Malayalam: Super Admin login cheyyumpol direct aayi central platform dashboard-ilekku access labhikkunnu.

Step 2: Urgent Approvals Banner

- English: Point out the Action Needed banner. Mention that cafeterias seeking plan renewal or card limit upgrades trigger real-time alerts here.
- Malayalam: Dashboard-il mukalilulla Action Needed alert kanikkuka. Plan renewal request vannal Super Admin-u ivide udan notification varum.

Step 3: 4 KPI Cards Scoping

- English: Demonstrate how selecting a specific cafeteria in the single dropdown changes all 4 cards (Cafeteria, Sales, Active Cards, Orders) to that exact cafeteria.
- Malayalam: Single dropdown-il oru cafeteria select cheythu 4 KPI card-ukalum (Sales, Cards, Orders) aa cafeteria-yude mathramayi maarunnath kanikkuka.

Step 4: Collapsible Directory

- English: Click "View Cafeterias" to expand the directory. Highlight the per-row action menu (Filter Dashboard, Manage Org, View Analytics, View Plans).
- Malayalam: View Cafeterias click cheythu table expand cheyyuka. Ooro row-ilumulla Action menu (Filter, Manage, Analytics) kanikkuka.

Step 5: Review & Approve Requests

- English: Navigate to Plans & Subscriptions -> Review Requests tab. Filter by status and click Approve to instantly update an organization's plan.
- Malayalam: Plans & Subscriptions-il Review Requests eduthu, filter cheythu, pending request single click-il Approve cheyyunnath demo cheyyuka.

Step 6: Live PDF Generation

- English: Open Platform Analytics. Click [View PDF] to preview the branded vector report in a modal, followed by [Download PDF].
- Malayalam: Analytics-il [View PDF] click cheythu in-browser preview-um, [Download PDF] vazhi direct download-um kanikkuka.

4. Expected Questions & Exception Scenarios (FAQ)

Q1: What happens if an organization exceeds its smart card limit or staff limit?

Ans (EN): The platform enforces real-time quota guardrails. When limit is hit, backend blocks new card issuance with 403 PLAN_LIMIT_EXCEEDED and prompts the Org Admin to request an upgrade.

Ans (ML): Card limit kazhinjal puthiya card issue cheyyan system sammathikkilla. Super Admin-u plan upgrade request ayakkendi varum.

Q2: How does the system handle offline POS transactions if internet fails in cafeteria?

Ans (EN): Cashier POS terminals use an encrypted localized transaction buffer with session tokens. When internet reconnects, all buffered transactions sync and reconcile automatically.

Ans (ML): Internet poyalum POS machine-il offline aayi bill cheyyam. Internet thirike varumpol automatic aayi central server-il sync aakum.

Q3: Can one cafeteria see sales or customer data of another cafeteria? (Security & Isolation)

Ans (EN): Strict multi-tenancy isolation is enforced at the database level. Every query and session token is scoped with organizationId. Only Super Admin has cross-tenant platform visibility.

Ans (ML): Oru cafeteria-kku mattulla cafeteria-yude sales-o data-yo kanikilla. Strict tenant isolation undu. Super Admin-u mathrame ellaam kananoo.

Q4: How are customer refunds processed and tracked by Super Admin?

Ans (EN): When a cashier initiates a customer refund, the balance is credited back to the customer wallet. The Super Admin dashboard tracks this under Customer Refunds and computes Net Revenue = Gross Purchases - Refunds.

Ans (ML): Oru customer-kku refund nalkumpol panam card-ilekku thirike credit aakunnu. Super Admin dashboard-il Net Revenue = Sales - Refunds aayi calculate cheyyunnu.

Q5: Can Super Admin set custom pricing or discounts for a specific VIP cafeteria?

Ans (EN): Yes. Super Admin can create Organization Overrides attached to a specific organizationId, granting custom pricing, extended billing, or elevated limits without changing public tiers.

Ans (ML): Athe, Super Admin-u oru prathyeka cafeteria-kku mathramayi custom plan override, discount, card limit enniva nalkan kazhiyum.

Q6: What prevents card balance tampering, cloning, or double-spending?

Ans (EN): Balances are not stored on plain NFC chips. They are managed on the secure cloud ledger. Tapping verifies encrypted UID and session nonce in real time, preventing cloning.

Ans (ML): Balance card-il alla, secure cloud ledger-il aanu sookshikkunnath. Card tap cheyyumpol server verify cheyyunnathukondu double-spending nadakkillaa.

Q7: How does client-side PDF export work without placing CPU load on the server?

Ans (EN): The PDF generation uses client-side jsPDF in the browser memory. It formats tables and charts vectorially without requiring backend headless browsers or rendering microservices.

Ans (ML): PDF generation poornamayi client browser-il (jsPDF) aanu nadakkunnath. Server bandwidth-o CPU load-o aavashyamilla.

Q8: How does order cancellation and top-up voiding work on the Mobile POS?

Ans (EN): Cashiers can cancel food orders directly from Food Orders or Action Hub, instantly refunding the card balance and restoring inventory. Top-ups can also be voided from Top-up History, deducting balance safely.

Ans (ML): Food orders cancel cheythai thuka udan card-ilekku refund aakum, stock restore cheyyum. Top-up history-il ninnu thettaya recharge void cheythu balance deduct cheyyam.

5. Technical Stack & Implementation Architecture

- Frontend Core: React 19, TypeScript, Tailwind CSS, Lucide Icons, Vite Fast Bundler
- PDF Vector Engine: Client-Side jsPDF Vector Engine with Dual-Mode (In-Browser Blob Preview & File Download)
- Backend Architecture: Node.js, Express, TypeScript, Prisma ORM, PostgreSQL Database
- Security & Tenant Isolation: Role-Based Access Control (RBAC), JWT Authentication, Strict Organization Scoping